Modelling the DC motor control system for EEE8089

Using GPL Maxima computer algebra system to model and design a control system for the use in EEE8089 coursework.

```
--> load(coma);
```

$coma v.1.84, (Wilhelm Haager, 2017-02-24)$

$$(\%o1) \quad /usr/share/maxima/5.41.0/share/contrib/coma/coma.mac$$

```
--> load(draw);
```

$$(\%o2) \quad /usr/share/maxima/5.41.0/share/draw/draw.lisp$$

For saving plots in EPS format add the following options to the plotting commands:

terminal=eps_color,dimensions=
[900,1000],line_width=2,file_name="plotname.eps"

For double plots choose 800x900, for single 800x450,

for high-resolution single column paper choose 1600x1800 or 1600x800,

for b/w terminal=eps, for web terminal=png

1 DC motor model

Modelling the DC motor as the first order low-pass stage.

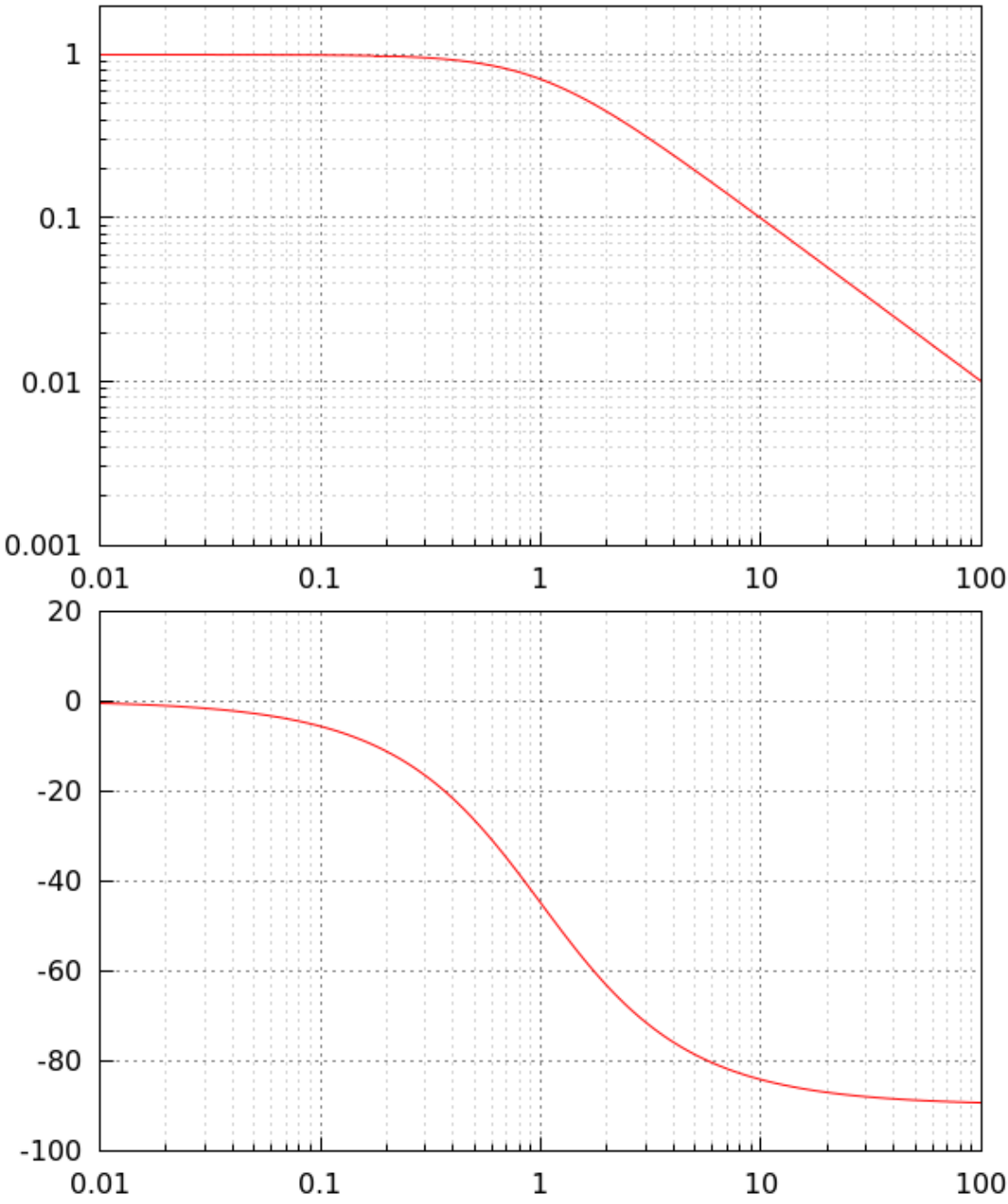
ODE: y'=1/tau*(x-y)

Laplace: K*1/(tau*s+1) or K*1/(s/(2*%pi*fc)+1);

The model below assumes K=1 and tau=1

```
--> bode_plot(1/(s+1),
      xrange=[0.01,100],yrange=[[0.001,2],[-100,20]]);
```

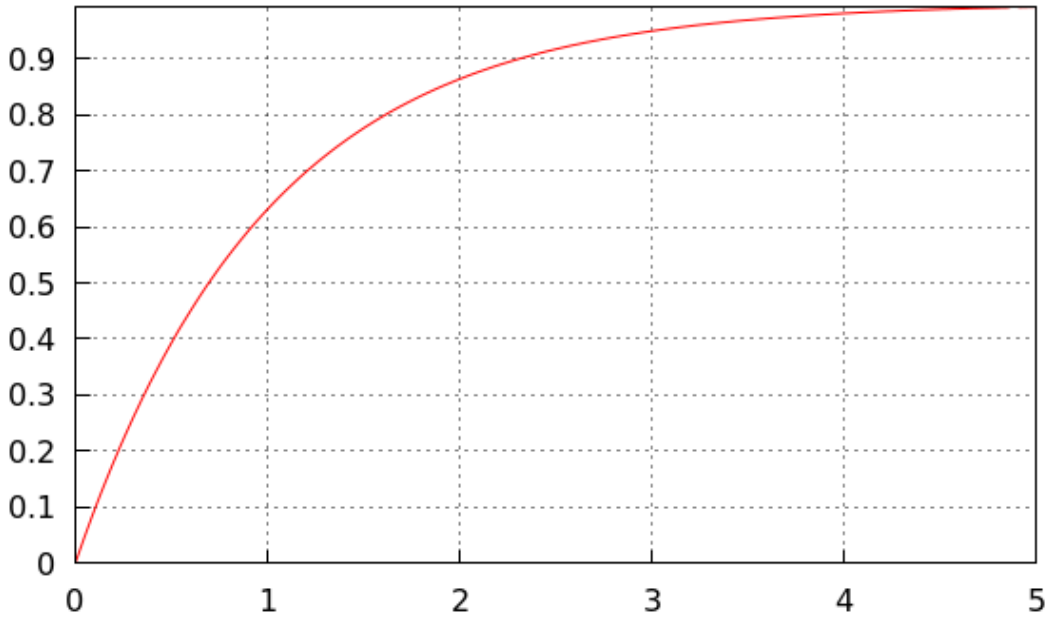
(%t3)



(%o3)

```
--> step_response([1/(s+1)],xrange=[0,5]);
```

(%t4)



(%o4)

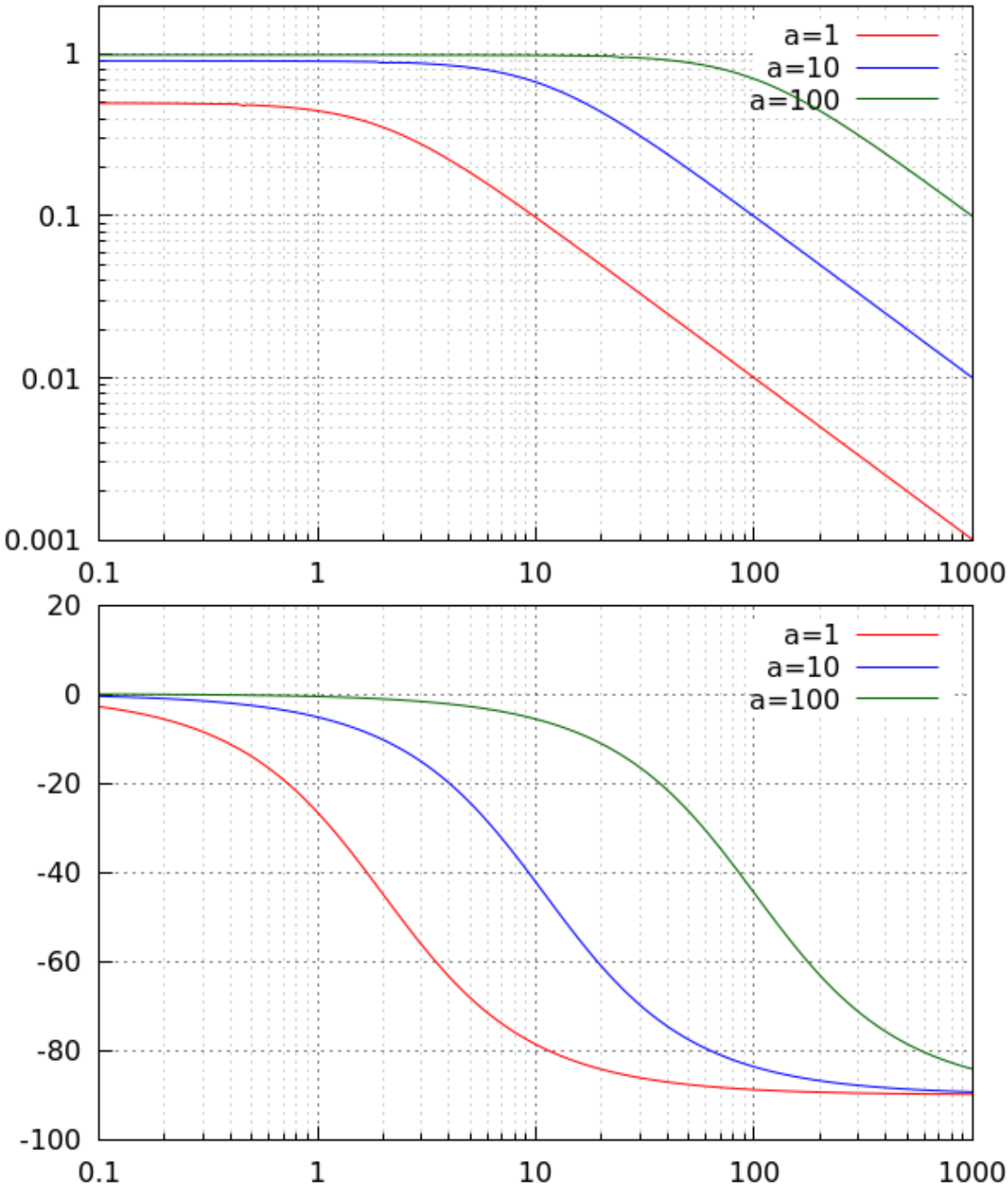
2 Ideal control loop

What if we include it in a closed control loop and add some gain 'a'?

Yes, it will work, but only if the loop does not include any processing delay, which is impossible...

```
--> bode_plot(create_list(closed_loop(a/(s+1)),a,[1,10,100]),
      xrange=[0.1,1000],yrange=[[0.001,2],[-100,20]],
      key=["a=1","a=10","a=100"]);
```

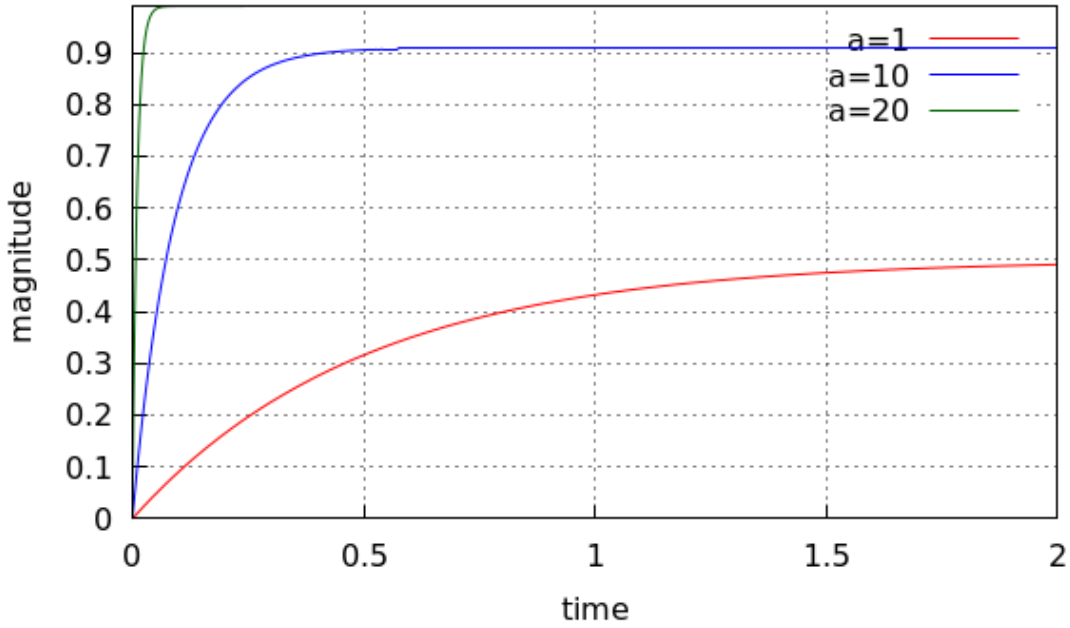
(%t5)



(%o5)

```
--> step_response(create_list(closed_loop(a/(s+1)),a,[1,10,100]),
      xrange=[0,2],xlabel="time",ylabel="magnitude",
      key=["a=1","a=10","a=20"]);
```

(%t6)



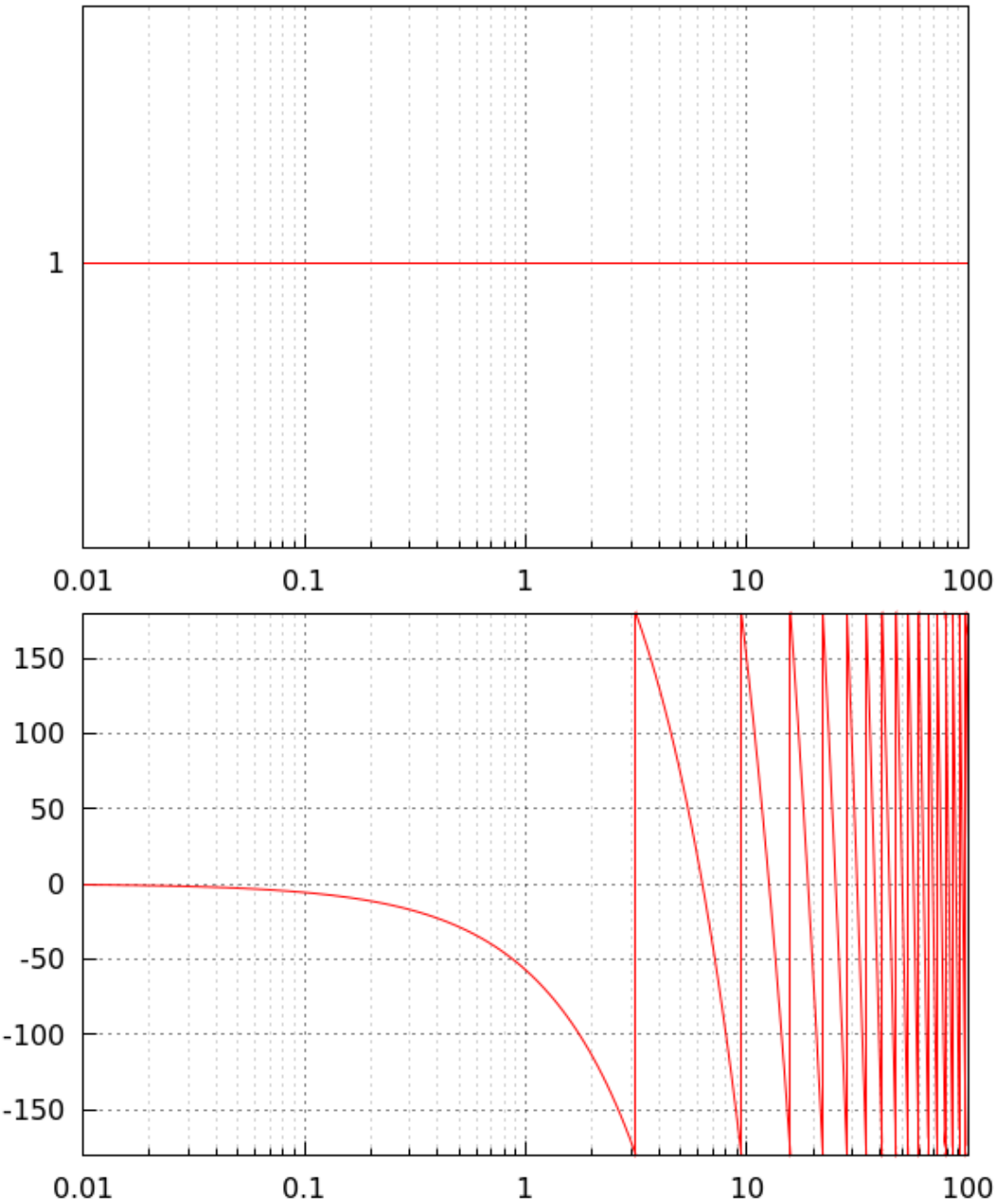
(%o6)

3 Delay model

The delay is caused by the driver board, frequency measurement and any communication between the blocks.

```
--> bode_plot(exp(-s),
      xrange=[0.01,100],yrange=[[0.9,1.1],[-180,180]]);
```

(%t7)



(%o7)

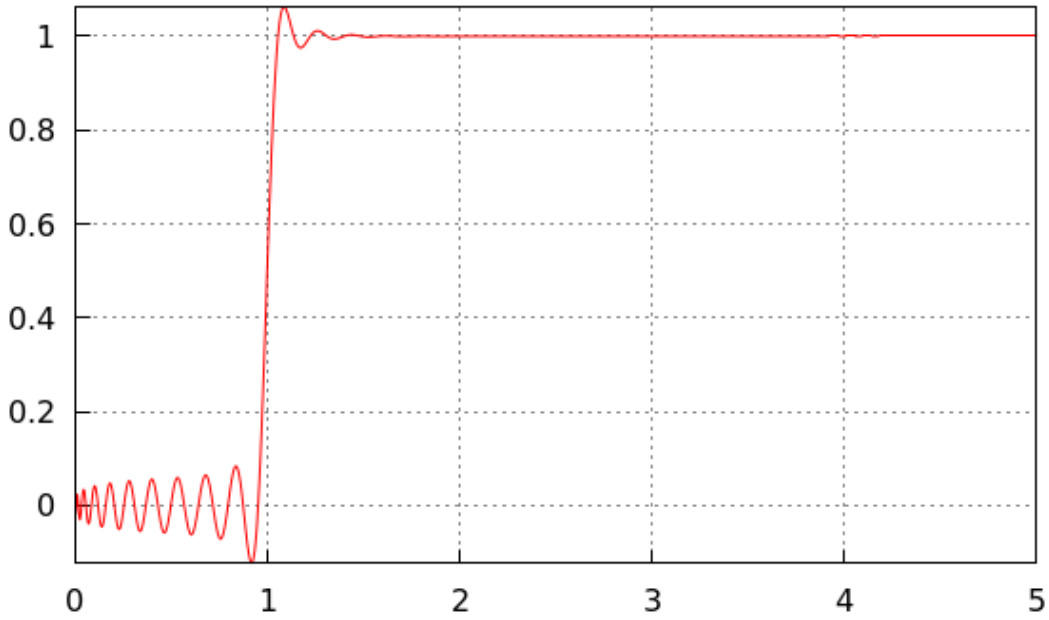
Just for fun, let's plot the step response of the delay line and then include it into a closed control loop.

It will oscillate if the gain is not less than 1.

exp(-a*s) has to be approximated with Pade series spelt as time_delay(a,approximation_order)

```
--> step_response(time_delay(1,20),xrange=[0,5]);
```

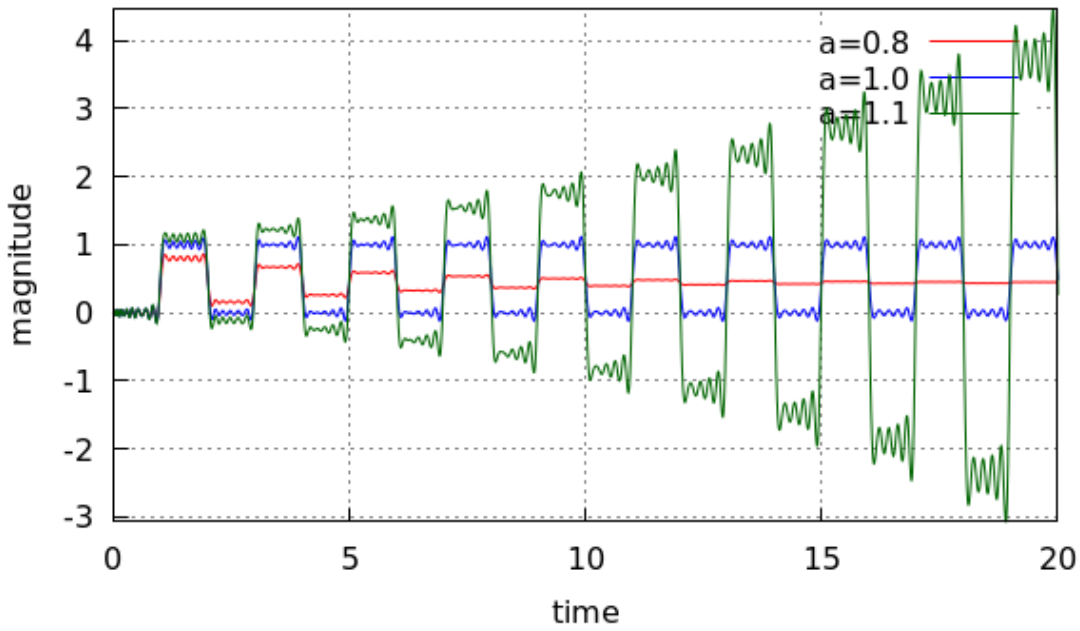
(%t8)



(%o8)

```
--> step_response(create_list(closed_loop(a·time_delay(1,20)),a,[0.8,1,1.1]),
      xrange=[0,20],xlabel="time",ylabel="magnitude",
      key=["a=0.8","a=1.0","a=1.1"]);
```

(%t9)



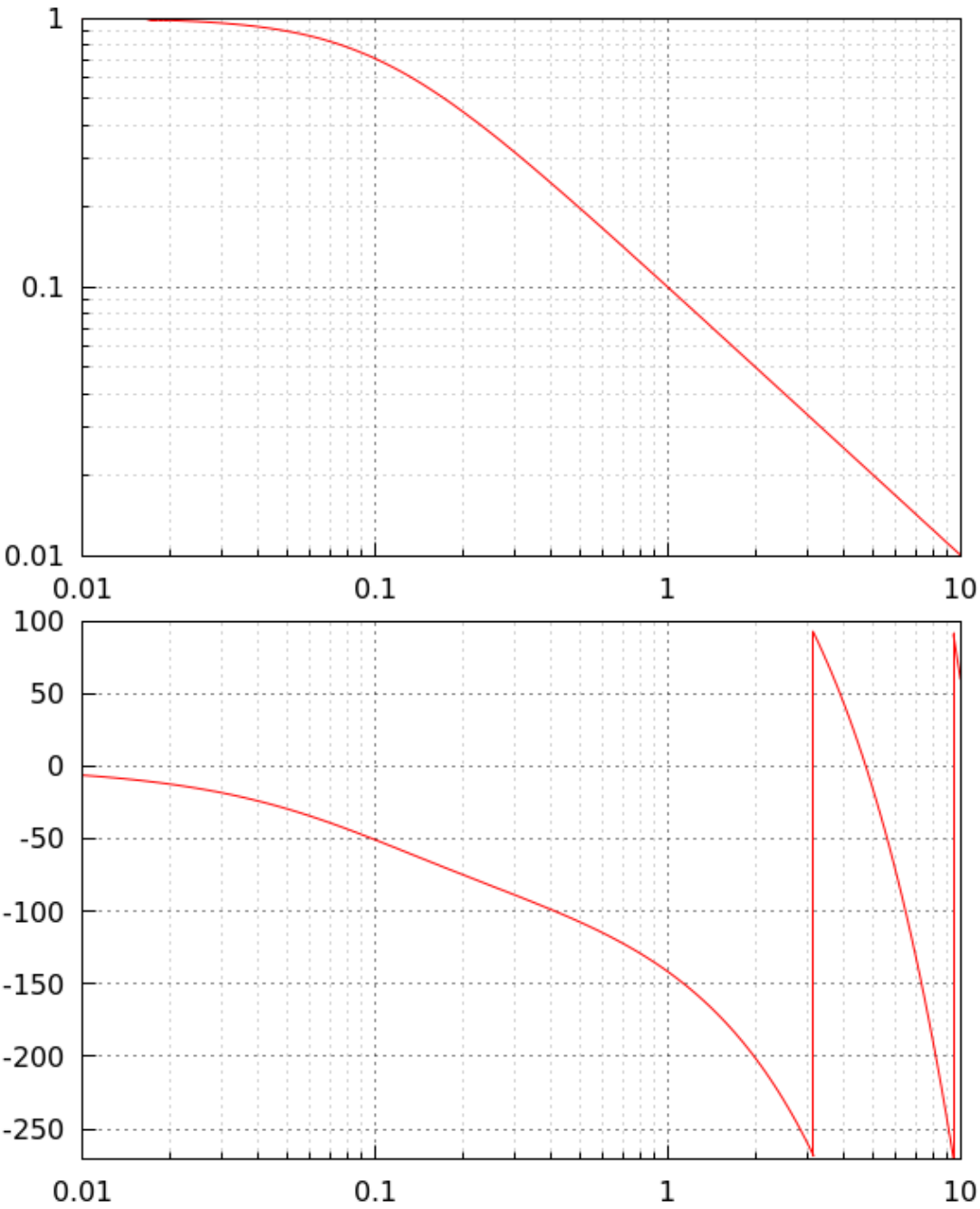
(%o9)

4 Motor*Delay

Assuming the time constant of the moto is 10x of the line delay.

```
--> bode_plot([exp(-s)·1/(10·s+1)],
      xrange=[0.01,10],yrange=[[0.01,1],[-270,100]]);
```

(%t10)



(%o10)

Let's try to include this into a control loop directly.

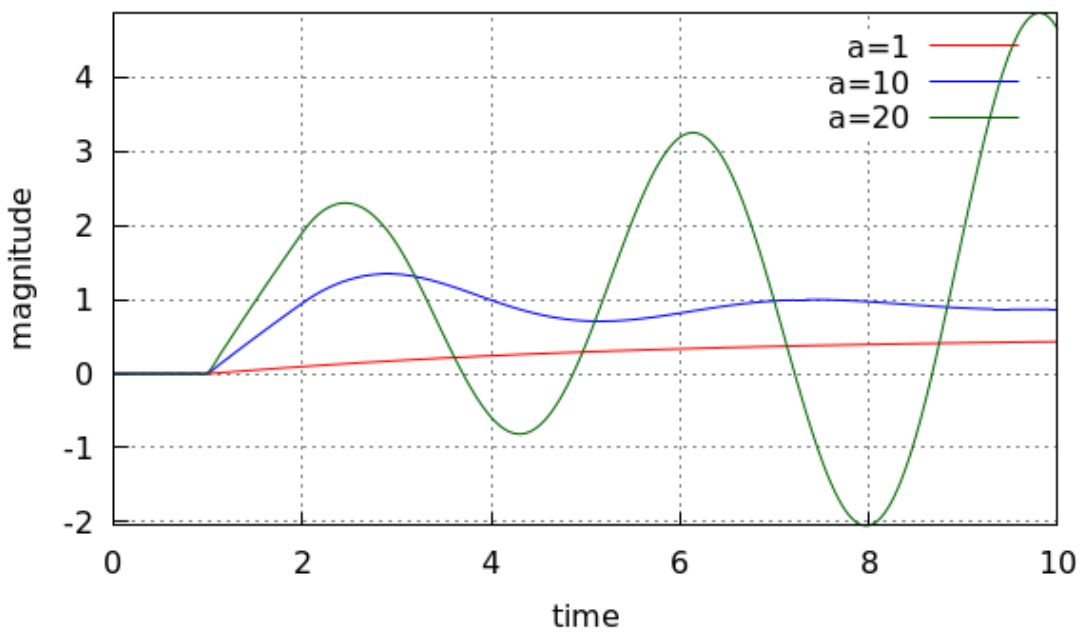
At -180 deg. phase the magnitude is about 1/17, which means that the maximum gain to add is 17.

Waht happens at different added gains 'a' is shown below; the system either oscillates,

or has a visibly large error. To increase gain at zero frequency one can use an integrator.

```
--> step_response(create_list(closed_loop(time_delay(1,20)·a/(10·s+1)),
        a,[1,10,20]),xrange=[0,10],xlabel="time",ylabel="magnitude",
        key=["a=1","a=10","a=20"]);
```

(%t11)



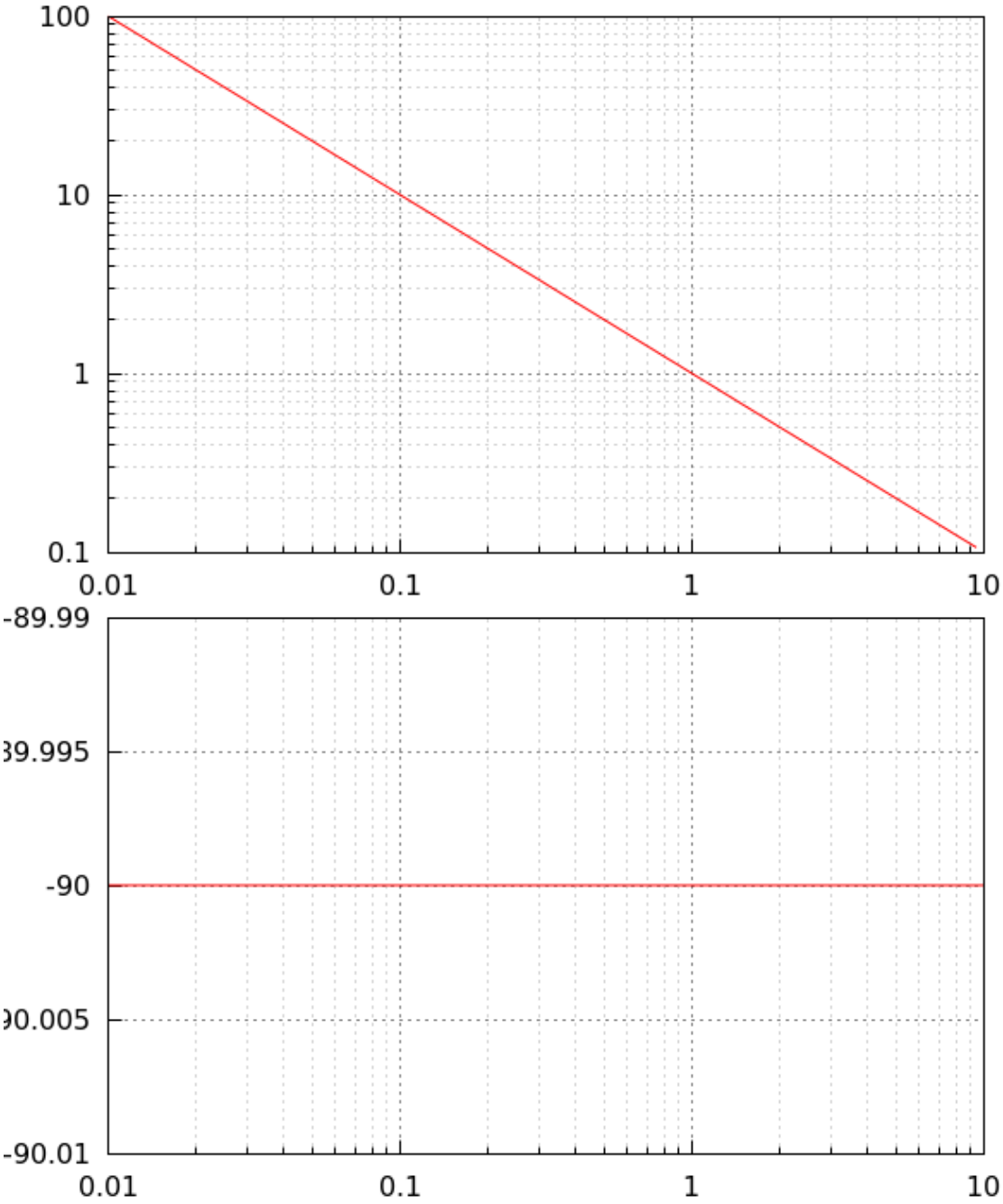
(%o11)

5 Integrator model

Integrators have infinite gain at frequency zero (static mode) and a constant phase shift of -90 deg.

```
--> bode_plot(1/s,
      xrange=[0.01,10]);
```

(%t12)



(%o12)

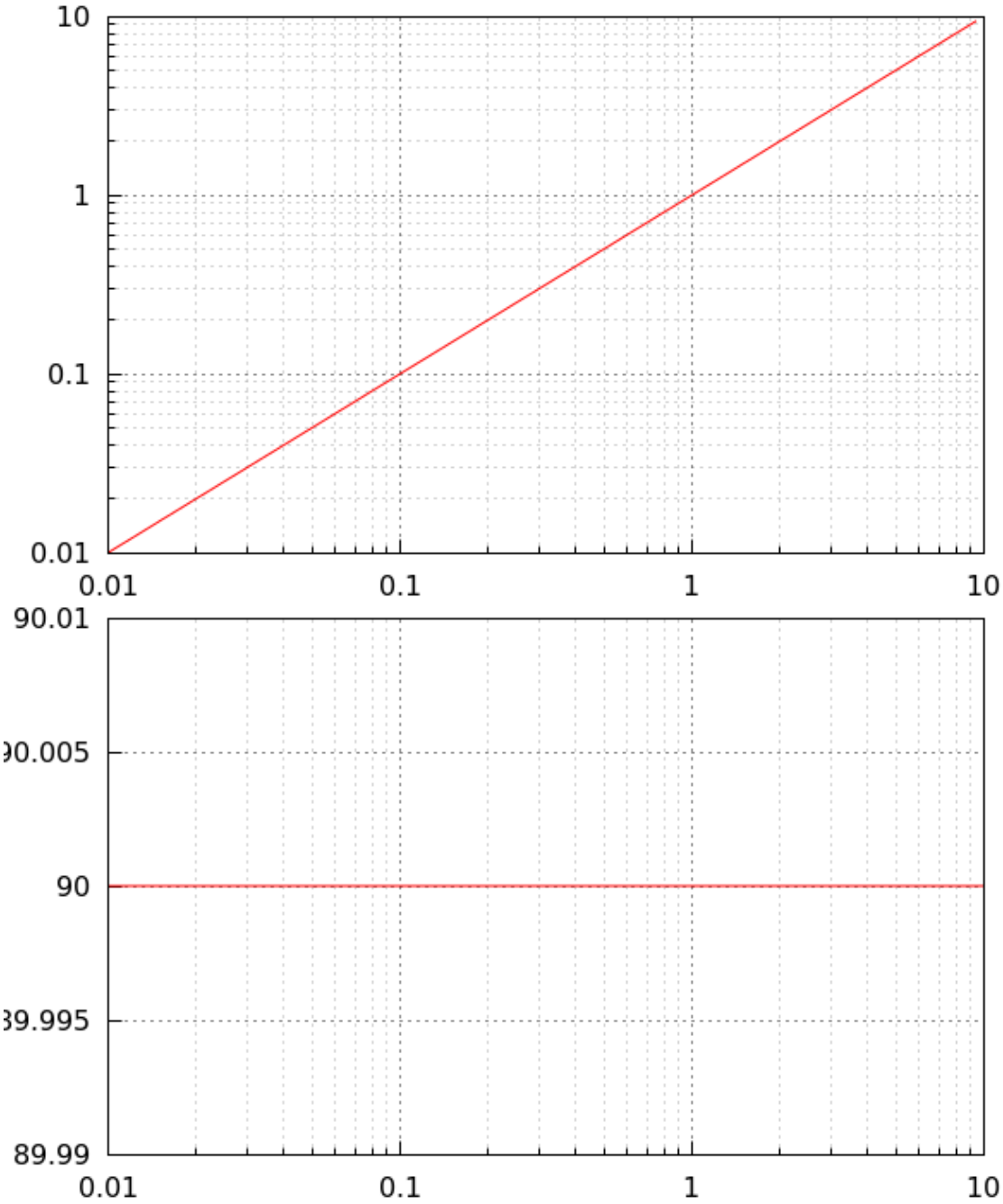
6 Differentiator model

Differentiators have zero gain at frequency zero (static mode), infinity gain at the infinity frequency

and a constant phase shift of +90 deg.

```
--> bode_plot(s,
      xrange=[0.01,10]);
```

(%t13)



(%o13)

7 Proportional-Integral control

Let's combine the earlier control system with an integrator and check

various combinations of the coefficients.

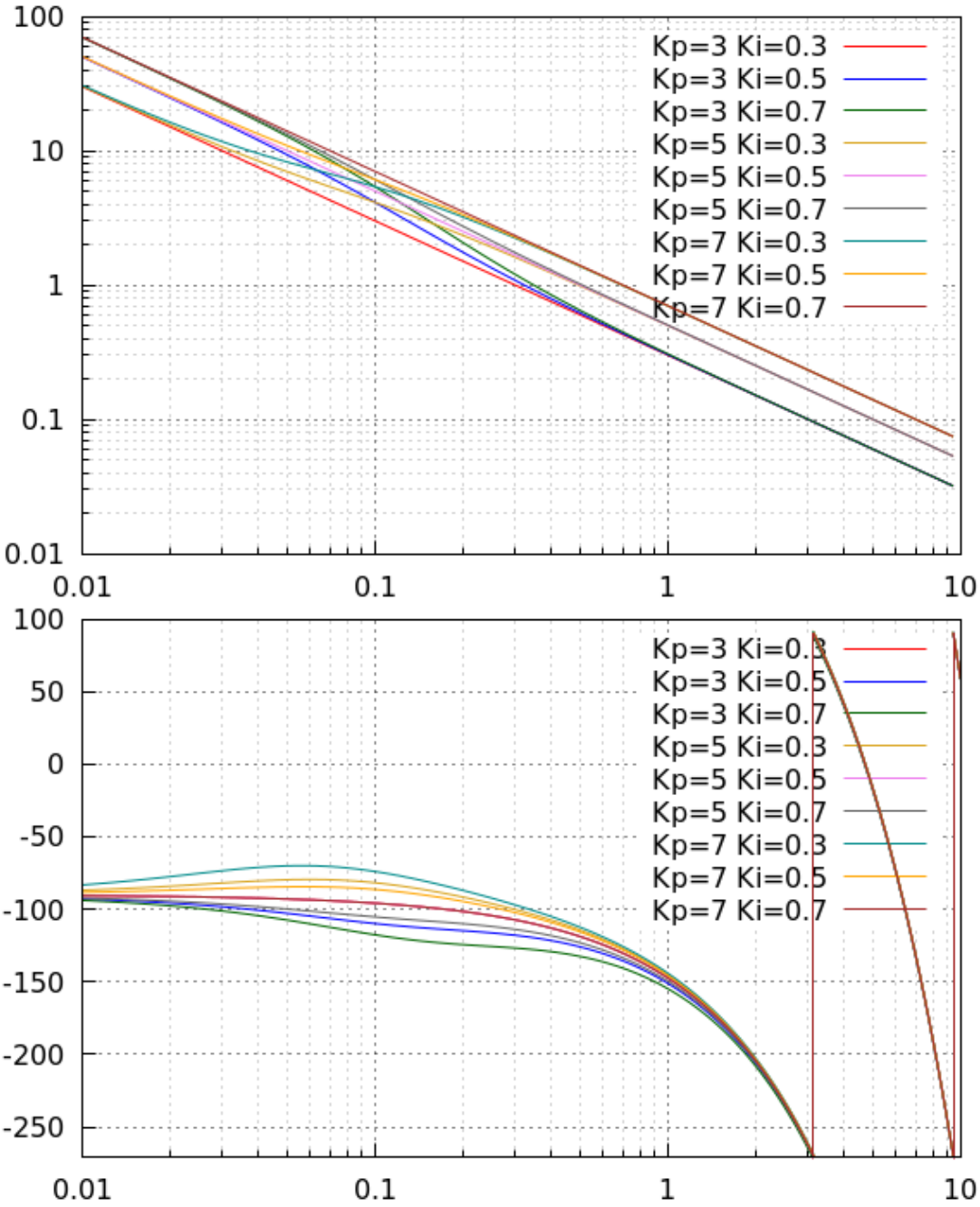
One can see that "Kp=5 Ki=0.5" option is the best, because it is stable, has low gain (-10dB) when the phase is -180 deg.,

and at the gain 0dB the phase shift is not far from -90 deg (about -110, which is within 20%).

Above 90 deg. overshoots begin, so itis good idea to not approach -180 deg.

```
--> bode_plot(create_list(exp(−s)·1/(10·s+1)·(Kp+Ki·1/s),Kp,[3,5,7],Ki,
    [0.3,0.5,0.7]),
      xrange=[0.01,10],yrange=[[0.01,100],[−270,100]],
      key=["Kp=3 Ki=0.3","Kp=3 Ki=0.5","Kp=3 Ki=0.7","Kp=5
    Ki=0.3","Kp=5 Ki=0.5","Kp=5 Ki=0.7","Kp=7 Ki=0.3","Kp=7
    Ki=0.5","Kp=7 Ki=0.7"]);
```

(%t14)

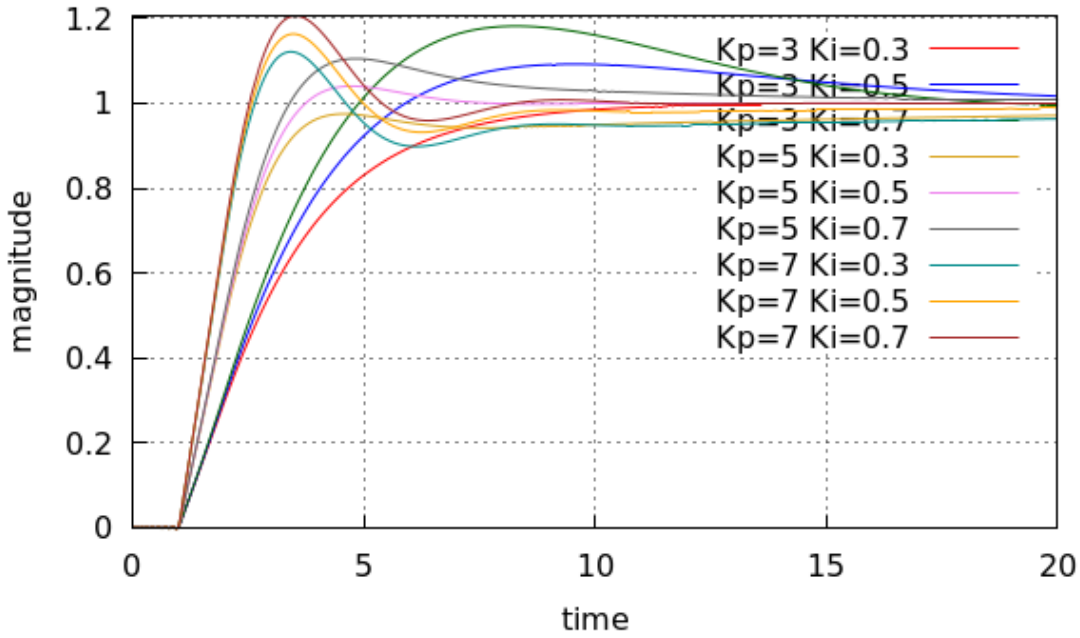


(%o14)

The above analysis is confirmed with step response diagrams below.

```
--> step_response(
      create_list(closed_loop(time_delay(1,20)·1/(10·s+1)·(Kp+Ki·1/s)),Kp,
    [3,5,7],Ki,[0.3,0.5,0.7]),
      xrange=[0,20],xlabel="time",ylabel="magnitude",
      key=["Kp=3 Ki=0.3","Kp=3 Ki=0.5","Kp=3 Ki=0.7","Kp=5
    Ki=0.3","Kp=5 Ki=0.5","Kp=5 Ki=0.7","Kp=7 Ki=0.3","Kp=7
    Ki=0.5","Kp=7 Ki=0.7"]);
```

(%t15)



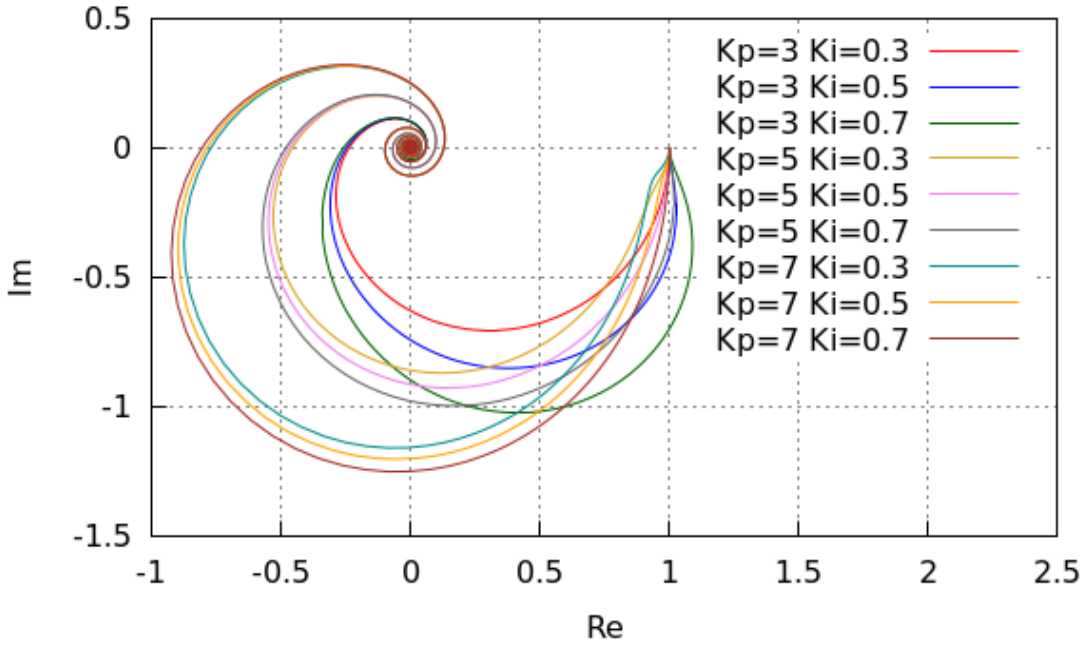
(%o15)

The same systems are shown in the Nyquist plot below.

This doesn't add much to the earlier observations, but is useful for completeness of knowledge.

```
--> nyquist_plot(create_list(
      closed_loop(time_delay(1,20)·1/(10·s+1)·(Kp+Ki·1/s)),Kp,[3,5,7],Ki,
    [0.3,0.5,0.7]),
      xrange=[−1,2.5],yrange=
    [−1.5,0.5],nticks=1000,xlabel="Re",ylabel="Im",
      key=["Kp=3 Ki=0.3","Kp=3 Ki=0.5","Kp=3 Ki=0.7","Kp=5
    Ki=0.3","Kp=5 Ki=0.5","Kp=5 Ki=0.7","Kp=7 Ki=0.3","Kp=7
    Ki=0.5","Kp=7 Ki=0.7"]);
```

(%t16)



(%o16)

8 Proportional-Integral-Derivative control

Let's combine the PI control system with a differentiator and check various combinations of the coefficients.

One can see that "Kp=7 Ki=0.7 Kd=2.2" option is the best, because it is stable, has low gain (-10dB) when the phase is -180 deg.,

and at the gain 0dB the phase shift is not far from -90 deg (about -110, which is within 20%).

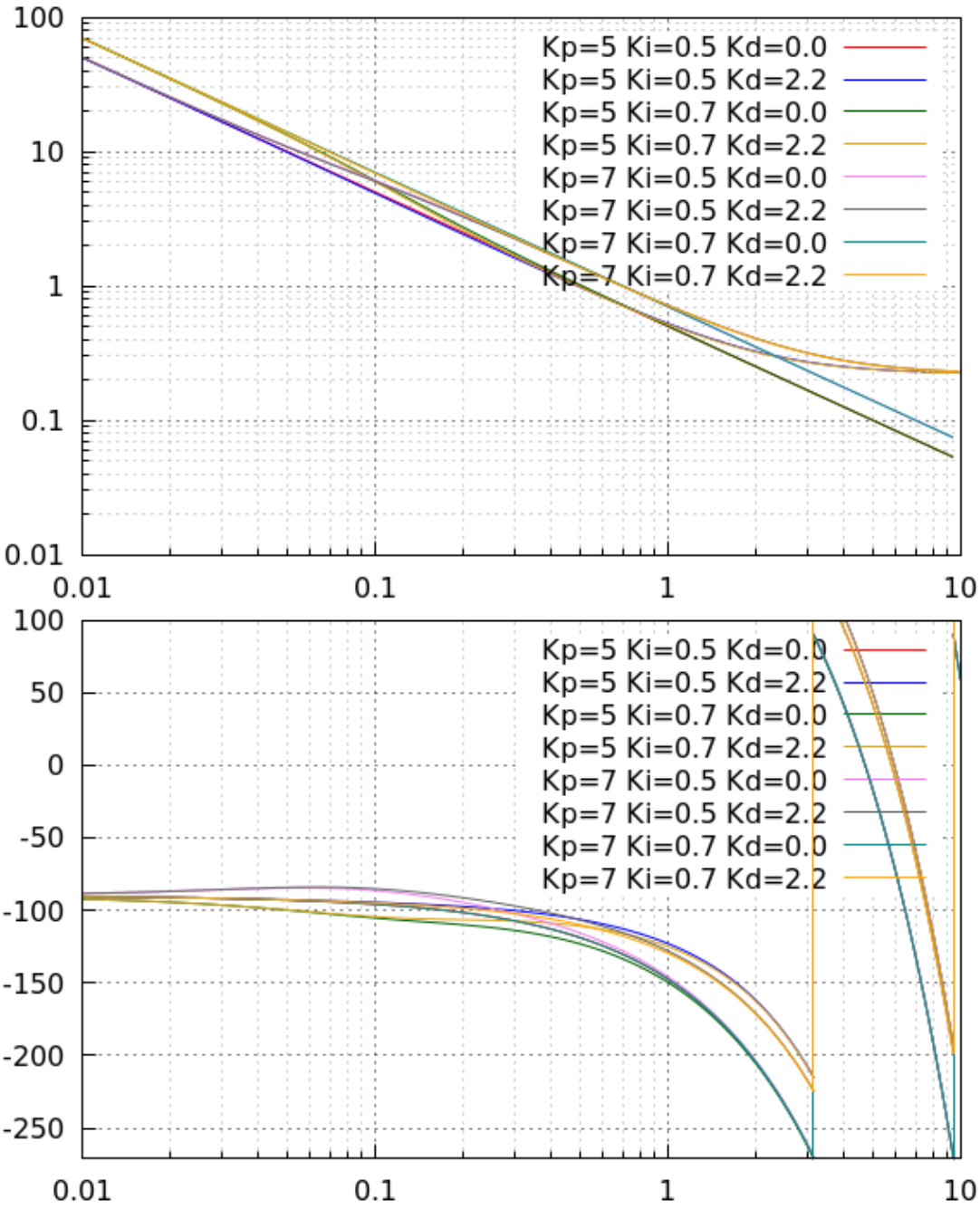
Above 90 deg. overshoots begin, so itis good idea to not approach -180 deg. Note, the unit gain takes place at a higher phase frequency

if compared to the PI controller.

```
--> bode_plot(create_list(exp(−s)·1/(10·s+1)·(Kp+Ki·1/s+Kd·s),Kp,[5,7],Ki,
    [0.5,0.7],Kd,[0,2.2]),
      xrange=[0.01,10],yrange=[[0.01,100],[−270,100]],
      key=["Kp=5 Ki=0.5 Kd=0.0","Kp=5 Ki=0.5 Kd=2.2","Kp=5 Ki=0.7
```

```
Kd=0.0","Kp=5 Ki=0.7 Kd=2.2",
      "Kp=7 Ki=0.5 Kd=0.0","Kp=7 Ki=0.5 Kd=2.2","Kp=7 Ki=0.7
Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t17)



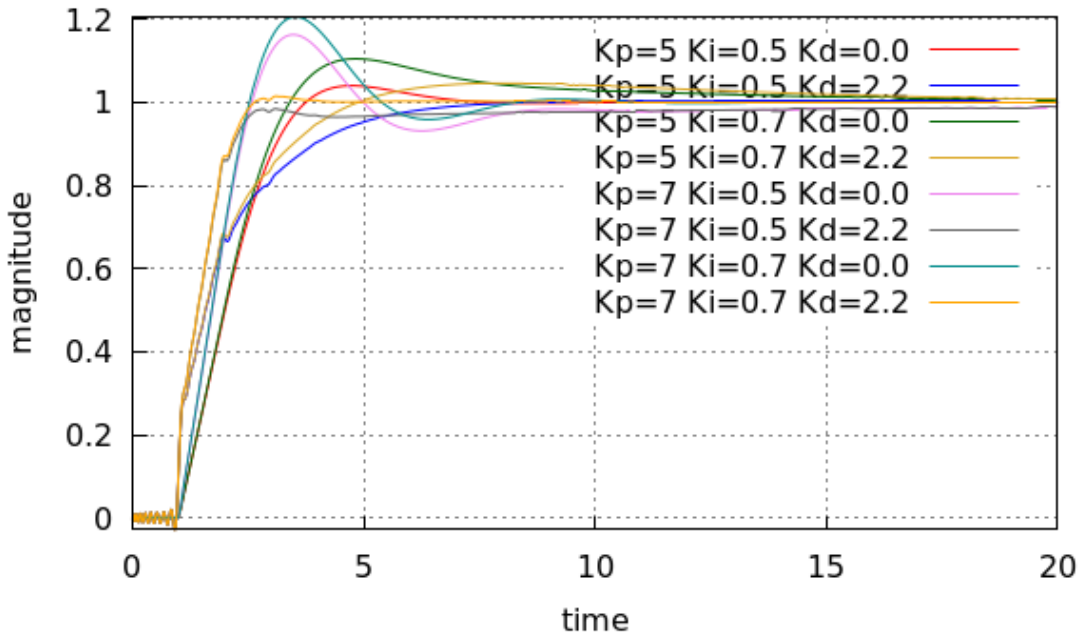
(%o17)

The above analysis is confirmed with step response diagrams below.

```
--> step_response(
      create_list(closed_loop(time_delay(1,20)·1/(10·s+1)·(Kp+Ki·1
```

```
/s+Kd·s)),Kp,[5,7],Ki,[0.5,0.7],Kd,[0,2.2]),
  xrange=[0,20],xlabel="time",ylabel="magnitude",
  key=["Kp=5 Ki=0.5 Kd=0.0","Kp=5 Ki=0.5 Kd=2.2","Kp=5 Ki=0.7
Kd=0.0","Kp=5 Ki=0.7 Kd=2.2",
      "Kp=7 Ki=0.5 Kd=0.0","Kp=7 Ki=0.5 Kd=2.2","Kp=7 Ki=0.7
Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t18)



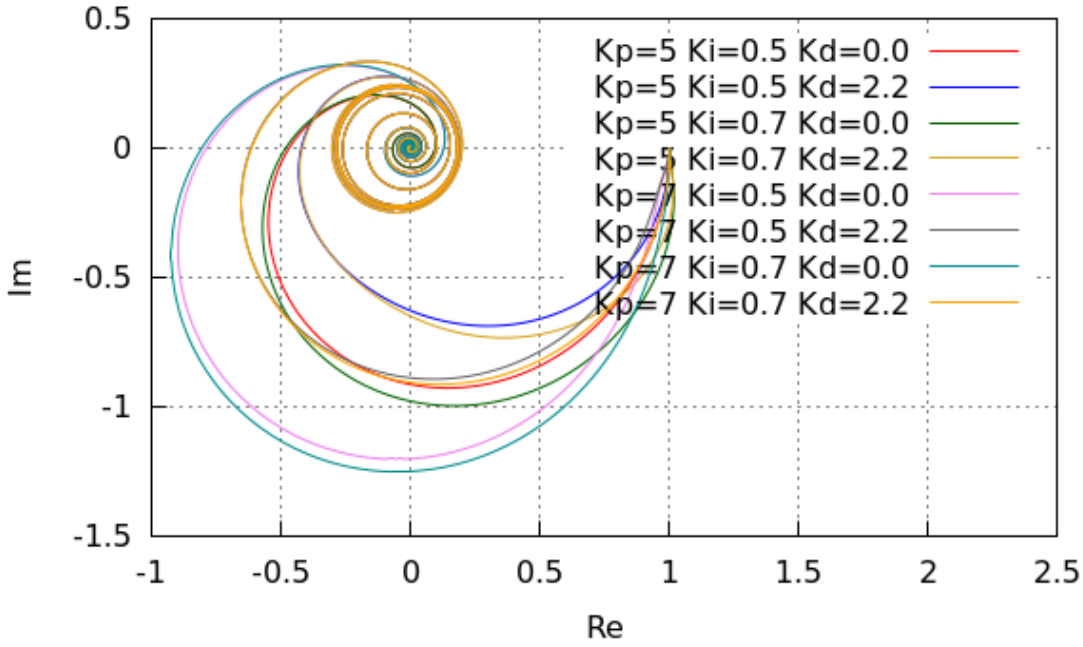
(%o18)

The same systems are shown in the Nyquist plot below.

This doesn't add much to the earlier observations, but is useful for completeness of knowledge.

```
--> nyquist_plot(create_list(
      closed_loop(time_delay(1,20)·1/(10·s+1)·(Kp+Ki·1/s+Kd·s)),Kp,
[5,7],Ki,[0.5,0.7],Kd,[0,2.2]),
      xrange=[−1,2.5],yrange=
[−1.5,0.5],nticks=8000,xlabel="Re",ylabel="Im",
      key=["Kp=5 Ki=0.5 Kd=0.0","Kp=5 Ki=0.5 Kd=2.2","Kp=5 Ki=0.7
Kd=0.0","Kp=5 Ki=0.7 Kd=2.2",
      "Kp=7 Ki=0.5 Kd=0.0","Kp=7 Ki=0.5 Kd=2.2","Kp=7 Ki=0.7
Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t19)

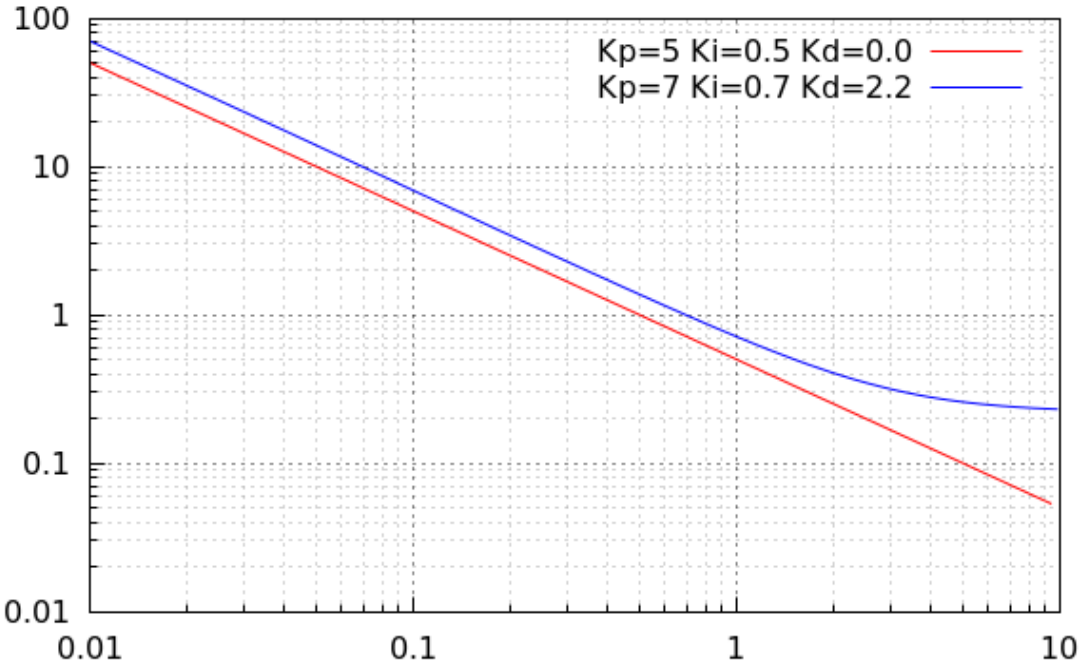


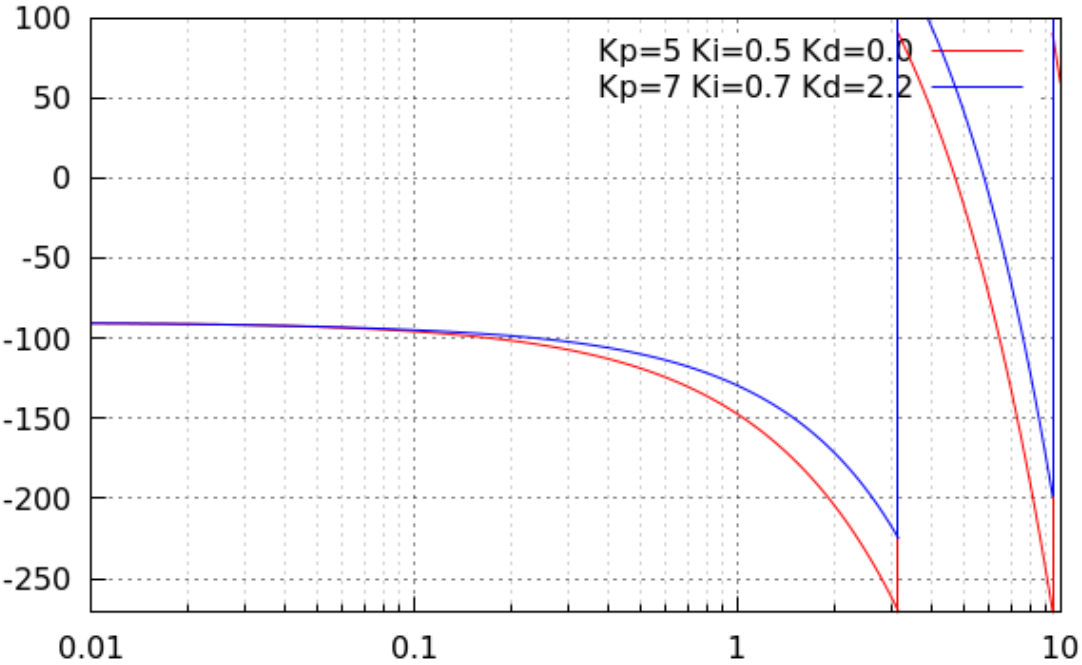
(%o19)

9 The best PI vs. PID comparison

```
--> bode_plot([exp(−s)·1/(10·s+1)·(5+0.5·1/s),exp(−s)·1/(10·s+1)·(7+0.7·1
    /s+2.2·s)],
      xrange=[0.01,10],yrange=[[0.01,100],[−270,100]],
      key=["Kp=5 Ki=0.5 Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t20)



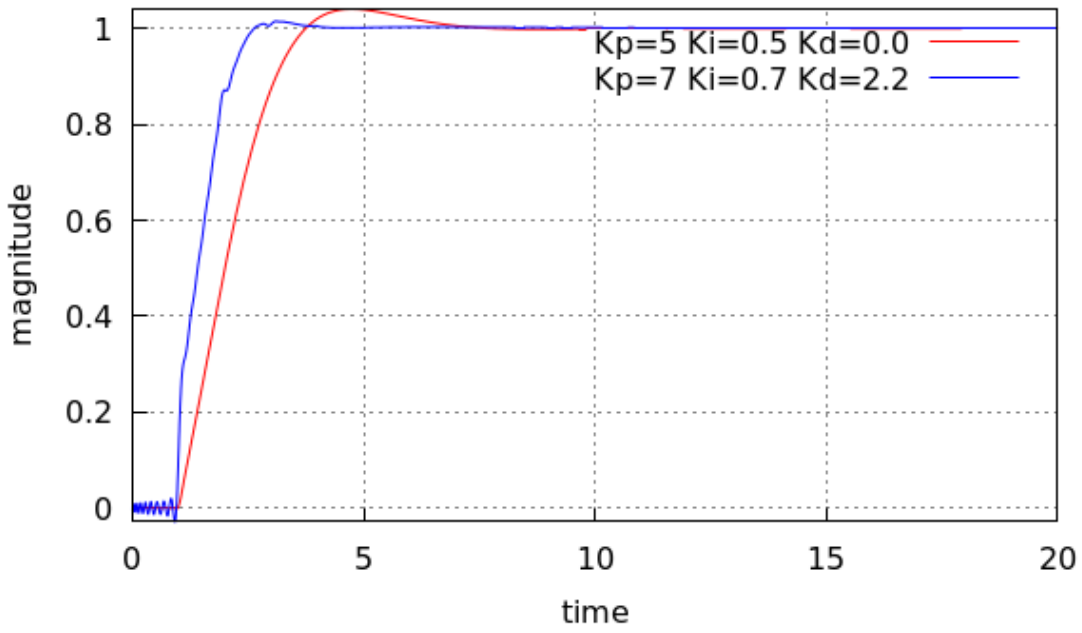


(%o20)

The above analysis is confirmed with step response diagrams below.

```
--> step_response([
        closed_loop(time_delay(1,20)·1/(10·s+1)·(5+0.5·1/s)),
        closed_loop(time_delay(1,20)·1/(10·s+1)·(7+0.7·1/s+2.2·s))],
      xrange=[0,20],xlabel="time",ylabel="magnitude",
      key=["Kp=5 Ki=0.5 Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t21)



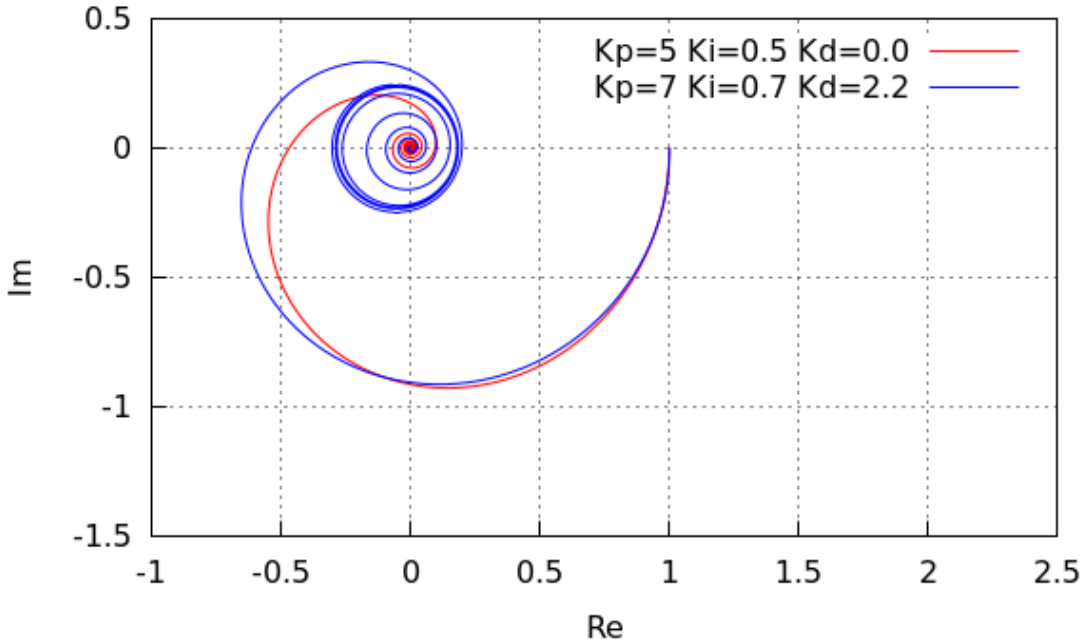
(%o21)

The same systems are shown in the Nyquist plot below.

This doesn't add much to the earlier observations, but is useful for completeness of knowledge.

```
--> nyquist_plot([
        closed_loop(time_delay(1,20)·1/(10·s+1)·(5+0.5·1/s)),
        closed_loop(time_delay(1,20)·1/(10·s+1)·(7+0.7·1/s+2.2·s))],
      xrange=[−1,2.5],yrange=
    [−1.5,0.5],nticks=8000,xlabel="Re",ylabel="Im",
      key=["Kp=5 Ki=0.5 Kd=0.0","Kp=7 Ki=0.7 Kd=2.2"]);
```

(%t22)



(%o22)

10 Conclusion

To summarise, the control loop has been designed (Kp, Ki and Kd found) for PI and PID controllers.

The tool Maxima is shown to be applicable to such designs.

Created with wxMaxima.